MATH 462 Topological Data Analysis through Persistent Homology

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Persistent homology is a method in topological data analysis to study the shape of data across many choices of scale. In ordinary homology, we study one fixed topological space and detect features such as connected components, loops, and higher dimensional voids. In topological data analysis, however, one usually begins with a finite cloud of sample points rather than a single canonical space. The main idea of persistent homology is to build a nested family of simplicial complexes from the data and then track which homology classes appear, survive, and disappear as the scale changes.

A simple motivating example comes from time series data. Consider the periodic sequence

$$1, 0, -1, 0, 1, 0, -1, 0, \dots.$$

Using a sliding window of length 2, and keeping the distinct resulting vectors, we obtain the vectors

$$(1, 0), \quad (0, -1), \quad (-1, 0), \quad (0, 1)$$

in $\mathbb{R}^2$. Thus a one dimensional time series has been converted into a finite metric space. If a time series has periodic or nearly periodic behavior, then a sliding window embedding may produce a point cloud with a loop like shape.

Definition 1. *Let $x_1, x_2, \dots$ be a sequence of real numbers and let $m \geq 1$. The sliding window embedding of length m is the collection of vectors*

$$(x_t, x_{t+1}, \dots, x_{t+m-1}) \in \mathbb{R}^m,$$

one for each valid index t.

Definition 2. *A filtration of simplicial complexes is a nested sequence*

$$K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_N.$$

Throughout this paper, homology is taken with coefficients in a field $\mathbb{F}$. For a simplicial complex K, the kth homology group is denoted by $H_k(K)$, and its dimension

$$\beta_k(K) := \dim_{\mathbb{F}} H_k(K)$$

is called the kth Betti number. Intuitively, β_0 counts connected components, β_1 counts independent loops, and higher Betti numbers count higher dimensional holes.

Since each inclusion $K_i \hookrightarrow K_j$ with $i \leq j$ induces a homomorphism $H_k(K_i) \to H_k(K_j)$, one can study not only the groups $H_k(K_i)$ themselves, but also how their classes survive through the filtration.

Definition 3. *Let $K_0 \subseteq K_1 \subseteq \cdots \subseteq K_N$ be a filtration. For $i \leq j$, the kth persistent homology group from stage i to stage j is*

$$H_k^{i,j} := \operatorname{im}\big(H_k(K_i) \to H_k(K_j)\big).$$

Also, we can note that its dimension $\beta_k^{i,j} := \dim_{\mathbb{F}} H_k^{i,j}$ is called the kth persistent Betti number.

Thus $H_k^{i,j}$ records the k dimensional homology classes present at stage i that are still present at stage j.

Definition 4. *Let X be a finite metric space and let $\varepsilon \geq 0$. The Vietoris–Rips complex $R_\varepsilon(X)$ is the simplicial complex whose vertices are the points of X, and for which a finite subset $\{x_0, \ldots, x_n\}$ spans an n simplex whenever*

$$d(x_r, x_s) \leq \varepsilon \qquad \text{for all } 0 \leq r < s \leq n.$$

In an intuitive sense, if we increase ε, then it would create more edges and that higher dimensional simplices would also appear.

Proposition 1. *For a finite metric space X, the family $\{R_\varepsilon(X)\}_{\varepsilon \geq 0}$ forms a filtration of simplicial complexes. In particular, $\forall \varepsilon$, $R_\varepsilon(X)$ is a simplicial complex, and if $\varepsilon \leq \varepsilon'$, then $R_\varepsilon(X) \subseteq R_{\varepsilon'}(X)$.*

Proof. Suppose $\sigma = \{x_0, \ldots, x_n\}$ is a simplex of $R_\varepsilon(X)$. So, every pair of vertices in σ has distance at most ε. Now note that any face of σ is obtained by deleting vertices, so every pair of vertices in that face also has distance at most ε. Therefore, every face of σ would then again be a simplex of $R_\varepsilon(X)$. Therefore $R_\varepsilon(X)$ is a simplicial complex.

Now suppose $\varepsilon \leq \varepsilon'$ and let σ be a simplex of $R_\varepsilon(X)$. Then every pair of vertices in σ has distance at most ε which would also mean at most ε'. Therefore σ is also a simplex of $R_{\varepsilon'}(X)$, so $R_\varepsilon(X) \subseteq R_{\varepsilon'}(X)$ which completes the proof. □

We now return to the four point example. Let

$$A = (1, 0), \quad B = (0, -1), \quad C = (-1, 0), \quad D = (0, 1).$$

With these points, the adjacent vertices would have a distance of $\sqrt{2}$, while the opposite vertices would have a distance 2. So, we know from here that the Vietoris–Rips complex would only change at these two values.

For instance, if $0 \leq \varepsilon < \sqrt{2}$, then no two distinct points will be joined by an edge. Hence $R_\varepsilon(X)$ consists of four isolated vertices, so

$$H_0(R_\varepsilon(X)) \cong \mathbb{F}^4, \qquad H_1(R_\varepsilon(X)) = 0.$$

Thus there are four connected components and no loop.

Now if $\sqrt{2} \leq \varepsilon < 2$, then adjacent vertices would become connected to each other, but opposite vertices are not. Hence $R_\varepsilon(X)$ becomes the boundary of a square. Therefore

$$H_0(R_\varepsilon(X)) \cong \mathbb{F}, \qquad H_1(R_\varepsilon(X)) \cong \mathbb{F}.$$

So the complex is now connected and one nontrivial loop appears.

For the next case, if $\varepsilon \geq 2$, then every pair of vertices has distance at most ε which would then cause all edges to be present such that every triple of vertices spans a triangle, and all four vertices span a 3 simplex. Now note that the complex is contractible because it is the full simplex on four vertices. So we have that;

$$H_0(R_\varepsilon(X)) \cong \mathbb{F}, \qquad H_1(R_\varepsilon(X)) = 0.$$

Definition 5. *A barcode for a fixed homological degree k is a collection of intervals, one interval for each persistent homology class. The left endpoint records when the class is born, and the right endpoint records when it dies.*

First, we shall use the convention that a class born at a and dying at b is represented by $[a, b)$. In degree 0, the four isolated vertices produce four bars at the beginning. At $\varepsilon = \sqrt{2}$, four connected components merge into one, so three bars die and one survives. Hence the degree 0 barcode is

$$[0, \infty), \quad [0, \sqrt{2}), \quad [0, \sqrt{2}), \quad [0, \sqrt{2}).$$

In degree 1, the loop is born at $\varepsilon = \sqrt{2}$ and dies at $\varepsilon = 2$. Hence the degree 1 barcode is

$$[\sqrt{2}, 2).$$

Therefore, the persistence length of this loop is then $2 - \sqrt{2}$ which concludes the example.

From here, we can see why persistent homology is more informative than ordinary homology, where in particular if we only chose a single value of ε, the answer could depend strongly on that choice. Whereas, persistent homology records the whole evolution of the topology. It does not by itself prove that a feature is useful in a particular application, but it does show whether the feature survives across a range of scales rather than appearing only at one arbitrary chosen parameter value.

In conclusion, persistent homology begins with a filtration of simplicial complexes, then applies homology at each stage, and finally studies the images of the induced maps between stages. The Vietoris–Rips construction gives a natural filtration for finite metric spaces. The barcode gives a compact summary of which connected components and loops persist across scale. In the four point time series example, a periodic sequence becomes a point cloud, the point cloud produces a Rips filtration, and the barcode records one connected component that persists forever together with one loop that exists only on the intermediate range $\sqrt{2} \leq \varepsilon < 2$.

References

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[2] G. Carlsson, *Topology and data*, Bulletin of the American Mathematical Society **46** (2009), 255–308.